Kudeki

Exam 1

Wed, June 28, 2017 — 11 AM - 11:50 AM

Name:	
Section:	

Please clearly PRINT your name in CAPITAL LETTERS and circle your section in the above boxes.

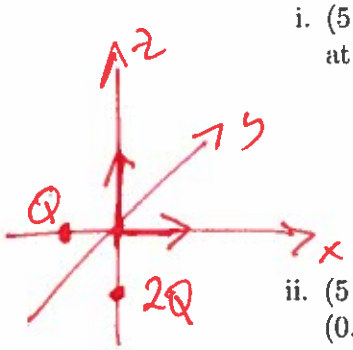
This is a closed book exam and calculators are not allowed. You are allowed one formula sheet of 8.5 by 11 inch dimensions — both sides of the sheet may be used. Please show all your work and make sure to include your reasoning for each answer. All answers should include units wherever appropriate.

Problem 1 (25 points)	
Problem 2 (25 points)	
Problem 3 (25 points)	
TOTAL (75 points)	

1. Two parts of this question are independent:

a) **Electrostatic fields of point charges:**

i. (5 pts) A point charge Q is located at $(x, y, z) = (-1, 0, 0)$ m and causes a field $\mathbf{E} = \hat{x}$ V/m at $(x, y, z) = (0, 0, 0)$. What is Q in Coulombs? — simplify your answer.



$\bar{E} = \frac{Q}{4\pi\varepsilon_0 r^2}\hat{r}$ $\Big|_{\substack{\hat{r}=\hat{x} \\ r=1\,m \\ Q=4\pi\varepsilon_0}} = \hat{x}\,\frac{V}{m} \Rightarrow Q = 4\pi\varepsilon_0 = \frac{4\pi}{36\pi\times10^9} = \frac{1}{9}\times10^{-9}$ C

ii. (5 pts) A $2Q$ charge is added at $(x, y, z) = (0, 0, Z)$ and the field direction at $(x, y, z) = (0, 0, 0)$ is changed to $\frac{\hat{x}+\hat{z}}{\sqrt{2}}$. What is Z in meters?

$E = \frac{2Q}{4\pi\varepsilon_0 r^2}\hat{z} = \hat{z} \Rightarrow \frac{2}{r^2} = 1 \Rightarrow r = \pm\sqrt{2}$

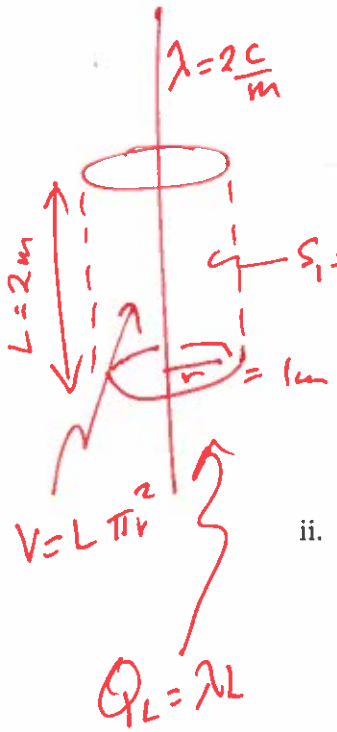
$\hookrightarrow Z_1 = -\sqrt{2}$ m.

iii. (5 pts) What is $|\mathbf{E}|$ in V/m at $(x, y, z) = (0, 0, 0)$ due to Q and $2Q$ in part (ii)?

$\bar{E} = \hat{x} + \hat{z} \Rightarrow |\bar{E}| = \sqrt{\bar{E}\cdot\bar{E}} = \sqrt{2}\ \frac{V}{m}$.

b) **Displacement flux and Gauss' Law:**

i. (5 pts) An infinite line charge with a $\lambda = 2$ C/m line density is located on z-axis surrounded by empty space. What would be the flux integral of displacement vector $\mathbf{D} = \epsilon_0\mathbf{E}$ over a cylindrical surface at a radial distance of $r = 1$ m from the z-axis if the length of the cylinder is $L = 2$ m (in z-axis direction)? — explain your reasoning.



$\oint_{S_1} \bar{D}\cdot d\bar{s} = Q_V = \lambda L = \left(2\frac{C}{m}\right)(2\,m) = 4C$

since $\oint_{caps} \bar{D}\cdot d\bar{s} = 0$

— flux calculated over S_1 only.

ii. (5 pts) Would the flux integral in (i) change if the line charge density outside the cylinder were zeroed out but remained $\lambda = 2$ C/m within the cylinder? — explain your reasoning and describe a change if any.

With $\lambda = 0$ outside the cylinder, we will have $\oint_{caps} \bar{D}\cdot d\bar{s} > 0$, and therefore $\oint_{S_1} \bar{D}\cdot d\bar{s} < 4C$

Yes, the flux of part (i) changes!

2. Three parts of the following question are independent:

a) Consider the electric field $\mathbf{E} = \mathbf{E}_1 + \mathbf{E}_2$, where $\mathbf{E}_1 = \hat{x}\cos(x)$ V/m and $\mathbf{E}_2 = \hat{x}\cos(y)$ V/m.

i. (4 pts) Which one of $\mathbf{E}_1$ and $\mathbf{E}_2$ satisfies the electrostatic field conditions? Explain.

$\nabla \times \bar{E}_1 = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ \cos(x) & 0 & 0 \end{vmatrix} = 0$, $\nabla \times \bar{E}_2 = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ \cos(y) & 0 & 0 \end{vmatrix} = \hat{z}\sin(y) \neq 0$ everywhere

$\bar{E}_1$ is electrostatic

ii. (4 pts) What is the charge density $\rho(x, y, z)$ in the region?

$\nabla \cdot \bar{E}_1 = -\sin(x) = \rho/\varepsilon_0$
$\nabla \cdot E_2 = 0$
$\Rightarrow \rho = -\varepsilon_0 \sin(x) \ \frac{C}{m^2}$

iii. (4 pts) What is the maximum value of "circulation per unit area" of $\mathbf{E}$ in the region? Explain your answer carefully. **Hint:** think of curl of $\mathbf{E}_1$ and $\mathbf{E}_2$.

$|\nabla \times E_2|_{max} = |\hat{z}\sin(y)|_{max} = 1 \ \frac{V}{m^2}$.

b) Given an electrostatic potential $V(x, y, z) = x^2y + 3$ V determine:

i. (3 pts) Electrostatic field $\mathbf{E}(x, y, z)$,

$\bar{E} = -\nabla V = -(2xy, x^2, 0) = -\hat{x}\,2xy - \hat{y}x^2 \ \frac{V}{m}$

ii. (3 pts) Charge density $\rho(x, y, z)$ in the region.

$\nabla \cdot \bar{E} = \frac{\partial E_x}{\partial x} + \frac{\partial E_y}{\partial y} = -2y = \frac{\rho}{\varepsilon_0} \rightarrow \rho = -2y\varepsilon_0 \ \left(\text{in } \frac{C}{m^3}\right)$ ($\partial E_y/\partial y = 0$)

c) (7 pts) Determine the electrostatic potential function $V(x, y, z)$ corresponding to $\mathbf{E} = y\hat{x} + x\hat{y} + \sin(z)\hat{z}$ if $V(0, 0, 0) = 3$ V.

$\bar{E} \cdot d\bar{l} = \underbrace{y\,dx + x\,dy}_{d(xy)} + \underbrace{\sin(z)\,dz}_{-d(\cos(z))} = -dV$

$d(xy - \cos(z)) = -dV$

$\rightarrow V = \cos(z) - xy + C$

$V(0,0,0) = 1 - 0 + C = 3 \Rightarrow C = 2$ $\rightarrow V(x, y, z) = \cos(z) - xy + 2$ V

3. Two parts of the following question are independent:

a) (16 pts) Consider the following spherically symmetric configuration of composite materials in steady-state equilibrium:

i. The region defined by $r \le a$, where $r = \sqrt{x^2+y^2+z^2}$ is the radial distance from the origin, has $\epsilon = \epsilon_o$, $\sigma = 10^6$ S/m, and holds a net charge of $Q = -2$ C.

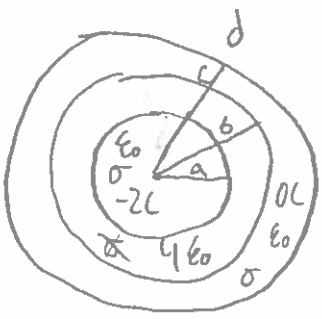
ii. A perfect dielectric shell with $\epsilon = 4\epsilon_o$ occupies the region $a < r < b$.

iii. Region $b \le r \le c$ has the same material properties as region $r \le a$ and holds zero net charge.

iv. Region $r > c$ is occupied by free space.In all four regions we have $\mu = \mu_o$.

Determine, in all four regions, (a) **D**, (b) **E** and (c) **P**, *and* (d) the surface charge densities, in C/m^2 units, at each of the three material boundaries at $r = a$, b, and c. **Hint:** Make use of Gauss's law in integral form, $\oint_S \mathbf{D}\cdot d\mathbf{S} = \int_V \rho dV$, with $\mathbf{D} = \epsilon\mathbf{E} = \epsilon_o\mathbf{E} + \mathbf{P}$, and a crucial fact about steady-state fields within conducting materials.

Very nice!

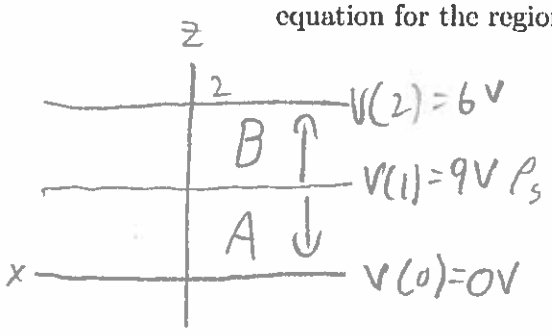


$E_a = \frac{-2}{4\pi\epsilon_0 4r^2}$ $\quad E_b = \frac{-2}{4\pi\epsilon_0 r^2}$

$\frac{4}{3\pi r^2} = \frac{-1}{8\pi^2} + P$

	$r < a$	$a \le r < b$	$b < r < c$	$c < r$
D =	0 C/m^2	$\frac{-\hat{r}}{2\pi r^2}$ C/m^2	0 C/m^2	$\frac{-\hat{r}}{2\pi r^2}$ C/m^2
E =	0 V/m	$\frac{-\hat{r}}{8\pi\epsilon_0 r^2}$ V/m	0 V/m	$\frac{-\hat{r}}{2\pi\epsilon_0 r^2}$ V/m
P =	0 C/m^2	$\frac{-3\hat{r}}{8\pi r^2}$ C/m^2	0 C/m^2	0 C/m^2
		$\rho_a = \frac{-1}{2\pi a^2}$ C/m^2	$\rho_b = \frac{1}{2\pi b^2}$ C/m^2	$\rho_c = \frac{-1}{2\pi c^2}$ C/m^2

b) (9 pts) Conducting plates placed on $z = 0$ and $z = 2$ m surfaces (of infinite extent) are separated by vacuum and maintained at potentials of $V(0) = 0$ and 6 V. Also, the potential of the $z = 1$ m surface that supports a surface charge density ρ_s is $V(1) = 9$ V. Determine ρ_s and the surface charge densities at $z = 0$ and $z = 2$ m by using the piecewise linear solution of the Laplace's equation for the region.



$\hat{n}\cdot D = \rho_s$

$V(z) = \begin{cases} 9z \text{ V}, & A \\ 9 - 3(z-1) \text{ V}, & B \end{cases}$

$E(z) = \begin{cases} -9\hat{z} \text{ V/m}, & A \\ +3\hat{z} \text{ V/m}, & B \end{cases}$

$D(z) = \begin{cases} -9\epsilon_0\hat{z} \text{ C/m}^2, & A \\ +3\epsilon_0\hat{z} \text{ C/m}^2, & B \end{cases}$

$\rho_{z=0} = -9\epsilon_0$ C/m^2

$\rho_{z=2} = -3\epsilon_0$ C/m^2

$\rho_s = \hat{z}\cdot(3\epsilon_0 + 9\epsilon_0)\hat{z}$

$\rho_s = 12\epsilon_0$ C/m^2